

Europe's first solar flowers?

Testing the Government's claims about Gozo's fifteen solar flowers against the record

THE CLAIM UNDER REVIEW

“With the installation of these first solar flowers in Europe, we are demonstrating how Gozo can be a model for sustainable innovation.”

Clint Camilleri, Minister for Gozo and Planning, 21 October 2025.

Lovin Malta's English; the Maltese original is in TVM News. Fifteen units at Ta' Xhajma, Gozo.

VERDICT: CONTRADICTED

Solar flowers were in service elsewhere in Europe from 2015

Version 1.0 · 6 October 2026

Status: draft, pending right of reply

Prepared from public sources, EU records and the PVGIS model.

Repository: github.com/leandergrech/Mizien

MIŻIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

On 21 October 2025 the Government announced fifteen “solar flowers” at the Gozo Multi-Modal Hub in Ta’ Xhajma, “the first of its kind in Europe” in TVM News’s report; Minister Clint Camilleri spoke of “these first solar flowers in Europe” and said their energy would supply the Park and Ride buses [1–3]. We checked both claims against the record.

- 1

Europe had solar flowers before Gozo. The units are SmartFlowers, an Austrian product [5]. Dated reports show them in Spain (2015), at an Austrian motorway rest area (2016) and at Vodafone’s UK headquarters (2021) [11–15]. The maker calls Gozo’s “one of the largest installations in Europe”, not the first [5].
- 2

The minister’s quote was narrower in one place. He also called it the first of its kind in Malta and, by number at one site, second in the world [1, 2]. Nothing we found contradicts the first; we did not test the second.
- 3

How much bus charging the flowers cover has not been shown. No capacity, metered output or bus use is published; Parliament was told no feasibility study was made [10] ♦. Modelled output, about 85 MWh a year, equals roughly 2–3 hours a day of the 10-minute shuttle (indicative) [17, 19, 23–25].
- 4

The tracking gain and the cost hold up. The Commission’s PVGIS model gives 36% more than fixed panels at the site, against “up to 40%” claimed [17]. About €850,000 of EU recovery funds fits the contract figures plus VAT [10, 26–28].
- 5

Verdict: contradicted (high confidence). Europe’s first solar flowers: contradicted by dated records. Powering the buses: not substantiated. Pending right of reply.

THIS CLAIM

▼

SUPPORTED

LARGELY SUPPORTED

NOT SUBSTANTIATED

MISLEADING

CONTRADICTED

2016

a SmartFlower was already in service at an Austrian motorway rest area [13]

15

SmartFlowers at Ta’ Xhajma; to the maker, “one of the largest installations in Europe”

≈85 MWh

a year modelled for 15 units of 2.5 kWp with two-axis tracking; none metered or published

2–3 h

of the 10-minute shuttle a day is what that output equals (indicative); bus use unpublished

WHAT WOULD MOVE THE VERDICT UP

Documents from October 2025 showing that “of its kind” meant a feature no earlier European installation had (for example the number of units at one site, or offsetting bus charging), and that this was said: *Misleading*, since “these first solar flowers in Europe” would still mislead.

WHAT WOULD MOVE IT DOWN

Nothing: *Contradicted* is already the lowest rating. Metered output and charging data would settle the bus-charging part, not this verdict.

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1

The claim and what we could verify

The Government’s own press release (gov.mt) could not be read from our network. Three readable texts carry it, and the claim is assembled from them. Only words inside quotation marks in those texts are treated as the speakers’ own; the rest is the outlet’s report of the statement. The Maltese article is the fullest: its English version drops the words “first ... in Europe” from the minister’s quote, while Lovin Malta’s English keeps them [1–3].

What was said	Who, where	Our access
“Bl-installazzjoni ta’ dawn l-ewwel solar flowers fl-Ewropa, qed nuru kif Ghawdex jista’ jkun ta’ eżempju fl-innovazzjoni sostenibbli.” In English: “With the installation of these first solar flowers in Europe, we are demonstrating how Gozo can be a model for sustainable innovation.”	Camilleri, quoted; TVM News (Maltese) [1]; Lovin Malta’s English [2]	Read in full
“... hija l-ewwel tax-xorta tagħha li qieghda tintuża f’pajjiżna u li bħala ammont fl-istess żona hija t-tieni wahda fid-dinja ...” (our translation: the first of its kind used in our country and, as a number in one place, the second in the world).	Camilleri, quoted [1]; Lovin Malta renders it “first of its kind in Malta” [2]	Read in full
“This energy will be used to supply electricity, electric buses which offer transport between Imġarr port and the Park & Ride facility.”	Camilleri, quoted; TVM News (English) [3]	Read in full
The installation is “the first of its kind in Europe”; the energy will directly offset the electricity used to charge the Park and Ride bus fleet; each flower yields up to 40% more than traditional panels; about €850,000.	The statement, in the outlets’ words, not quoted [1–3]	Read in full
“They are the first of their kind in Europe.” Funded through RRP C1-I5; €850,000.	European Commission project page, its own words [4]	Read in full
Parliamentary Secretary Omar Farrugia: a solar system “unika fl-Ewropa” (“unique ... in Europe”).	Quoted [1, 2]	Read in full

FAIRNESS NOTE

The flowers exist, work and were paid for with EU recovery funds, as the Government said; Parliament was told all fifteen were in working order [10] ♦. The minister’s own words also contain the narrower and accurate-looking claim that this is a first for Malta. The Commission page repeats the Government’s description; it does not test it. This check is about the words used, not about whether the project was worth doing.

2

Method

Questions. (A) Were these Europe’s first solar flowers, or the first installation of their kind, and what else could “first” mean? (B) How much electricity can fifteen flowers make, and how does that compare with what the Park and Ride buses use? (C) Is the stated tracking gain plausible? (D) What did the project cost and who paid?

Evidence. (A) The maker’s pages [5–8] and dated reports of earlier European installations [11–16]; search log in *literature/CC-032/notes.md*. (B, C) The European Commission’s PVGIS 5.3 model for the site [17], the data sheet [8, 9], GRDA and EU procurement records for the shuttle [19, 20], OpenStreetMap distances [23], and peer-reviewed estimates of bus consumption [24, 25] and tracking gains [18], read as abstracts. (D) Newsbook’s report of answers in Parliament [10] ♦, an EU award notice [26], the EU VAT table [27] and the Council decision on Malta’s recovery plan [28]. Inputs are in *data/cc-032/*; every figure is recomputed by *tools/cc-032-report/calc.py*.

Grades. Individual news reports and company pages are grade D, but the records of earlier installations are independent of one another, dated, and from three countries. PVGIS, EU procurement records, the Council decision and the GRDA are grade C. The tracker study [18] uses measured radiation across a latitude gradient (grade B); the bus-consumption figures are modelled for other routes and buses (grade C, indicative here). ♦ marks second-hand

material. **Verdicts** follow the five-point scale in Appendix A.

3 What a solar flower is

A SmartFlower is a free-standing photovoltaic unit with twelve petal-shaped panels that fan open at sunrise, follow the sun on two axes and fold away at night or in strong wind [4, 7–9]. The maker gives 2.5 kWp and 4,000–6,500 kWh a year per unit, depending on location [8, 9]. The product is manufactured in Austria, where the company was founded; a Boston firm bought it in 2018 [7].

Tracking raises output because the panels face the sun for more of the day. A study of five tracker types at locations across Europe and Africa found dual-axis trackers collected 17.7% to 31.2% more solar energy than panels fixed at the best angle, the gain varying with latitude [18]. The maker and the Government both say “up to 40%” [3, 7]; the PVGIS model gives 36.5% for this site (Section 5), but no measurements of these units are published, and the petal design may not behave like an ideal tracker.

The capacity of the Gozo units has not been published in anything we read. If they are the 2.5 kWp model the maker sells, the fifteen add up to 37.5 kWp; we use that figure below and say so each time.

4 Were they the first in Europe?

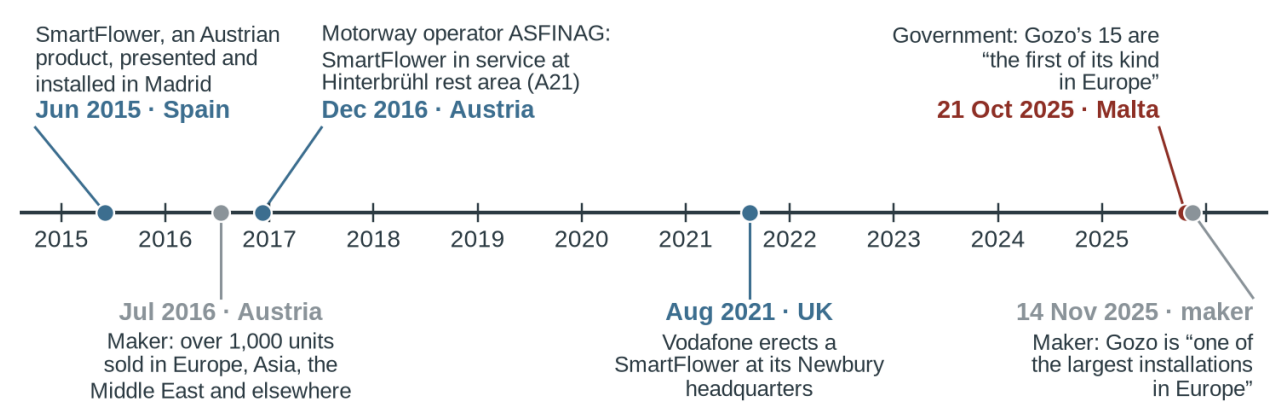


Figure 1. Dated records of SmartFlowers in Europe before the Gozo statement, and the maker's own description of Gozo's installation afterwards [5, 11–13, 15]. Records listed in data/cc-032/european_records.csv.

Date	Where	What the record says	Source
Jun 2015	Madrid, Spain	Smartflower, “made in Austria”, presented and installed in Spain; 3,400–6,000 kWh a year depending on site	energynews.es [11]
Jul 2016	Vienna-based maker	“Austria’s SmartFlower”; the company had sold over 1,000 units in Europe, Asia, the Middle East and elsewhere (its own figure)	pv magazine [12]
Dec 2016	Hinterbrühl rest area, A21, Austria	Motorway operator ASFINAG puts a SmartFlower into service; it covers half of the rest area’s lighting electricity, about 3,550 of 7,000 kWh a year	MeinBezirk.at [13]; IÖB [14]
Aug 2021	Newbury, UK	Vodafone erects a SmartFlower at its headquarters, made in Austria and installed by a UK distributor	Newbury Today [15]
Undated	UK, Jersey, Denmark, Spain	Dealers list 17 UK and Jersey installations, some of two or three units, and clients in Denmark, Spain and Scotland	Dealer pages [16]

Date	Where	What the record says	Source
14 Nov 2025	Ta' Xhajma, Gozo	Maker: the fifteen SmartFlowers are “one of the largest installations in Europe”; its year review repeats this	SmartFlower [5, 6]

Reading across the records

- **As solar flowers, not the first.** The same product was in service in Austria nine years earlier, including at a motorway rest area run by the Austrian motorway operator ASFINAG, and in the UK four years earlier [13–15].
- **As an installation of this size, not shown.** The maker calls it “one of the largest installations in Europe” [5]; it does not call it the largest. The maker’s English version of the minister’s quote reads “By installing these [SmartFlowers] in Europe”, with the word “first” replaced [5].
- **As a first for Malta, plausible.** Parliament was told no other such installation exists in Malta [10] ♦, and our web search found none. The minister’s claim to be second in the world by number at one site was not tested.
- **As a use, not unique.** Offsetting bus charging at a transport hub may be new; nothing we found shows an earlier European flower array charging buses. But the Government did not say that this was the novelty, and the 2016 Austrian unit already supplied part of a motorway rest area’s lighting.

5 Do they power the buses?

The minister said the energy “will be used to supply electricity, electric buses” running between Mġarr and the Park and Ride [3]; the statement said it would “directly offset” the electricity used to charge that fleet [2]. Neither gives an amount. Six electric buses owned by the Ministry for Gozo run the shuttle every 10 minutes, operated by Malta Public Transport under negotiated contracts [19, 20]. The GRDA notes that uptake “remains very low” [19]. Since May 2026 the hub has also held a new depot where Gozo’s electric buses, now 29 in all, are charged during the night [21, 22]; Malta Public Transport says its electric buses charge “primarily overnight” [21].

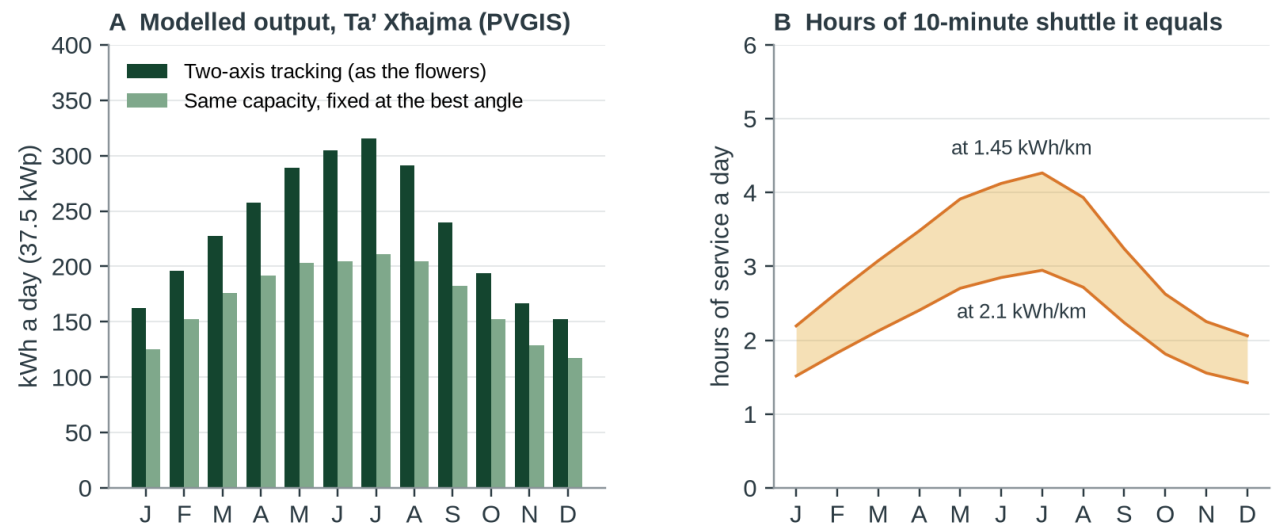


Figure 2. A: modelled output a day by month for 37.5 kWp at the Park and Ride, with two-axis tracking and with fixed panels at the best angle (PVGIS 5.3, 14% system loss) [17]. B: the hours of a 10-minute shuttle (8.5 km round trip [23]) that each month’s average day equals, at 1.45–2.1 kWh per km, figures modelled for other buses and routes [24, 25]. Indicative only: the Gozo buses’ consumption and hours are not published.

Indicator	Value	Source
Capacity of the 15 units	Not published; 37.5 kWp if 2.5 kWp each	[8, 9]
Modelled output, two-axis tracking	85,121 kWh a year ; 5,675 per unit; 233 kWh a day on average (152 in December, 315 in July)	[17], calculated
Maker's range for 15 units	60,000–97,500 kWh a year	[8], calculated
Same capacity, fixed at the best angle (32°)	62,341 kWh a year: tracking adds 36.5%	[17], calculated
Metered output of the flowers	Not published	Search log
Shuttle: buses, frequency, round trip	6 buses, every 10 minutes, 8.5 km: about 51 km per hour of service	[19, 20, 23]
Shuttle buses' model, battery, consumption, hours	Not published	Search log
Energy per hour of 10-minute service at 1.45–2.1 kWh/km	74–107 kWh (indicative)	[24, 25], calculated
Hours of that service the flowers' average day equals	2.2–3.2 hours ; 1.4–2.1 in December (indicative)	calculated

All inputs in data/cc-032/; results in checks.csv.

Three things follow. First, the amount of charging the flowers cover cannot be calculated from public data: neither their output nor the buses' consumption has been published, and Parliament was told that no technical report, feasibility study or value-for-money comparison was made [10] ♦. Second, on the only sourced figures we could find, the flowers' average day matches roughly two to three hours of the 10-minute shuttle, less in winter; whether that is all or part of its charging depends on hours and consumption that are not public. Third, the flowers produce only in daylight. They can feed daytime charging or other loads at the hub directly, and offset night-time charging on a yearly balance through the grid, which is what the statement's "offset" means; they cannot directly power buses charged overnight without storage, and none is mentioned. How they are wired is not published.

6

Where the evidence points different ways

Q1 Are they the first of their kind in Europe?		NO, AS WORDED
EVIDENCE FOR THE CLAIM'S IMPRESSION Gozo's array of fifteen is large: the maker calls it one of Europe's largest [5]. Using flowers to offset bus charging at a transport hub may be new; we found no earlier European example. It is the first in Malta as far as we could find [10] ♦.	EVIDENCE AGAINST The same product was installed in Spain (2015), Austria (2016) and the UK (2021), among others [11–16]. The maker does not call Gozo's the first, or the largest [5, 6]. The minister said "these first solar flowers in Europe" [1, 2], and the Commission page repeats it [4].	
WHY THEY DIFFER, AND WHAT IT MEANS HERE For this claim: as worded it is contradicted. A narrower claim (first in Malta, or among the largest in Europe) would have been supported by what we found.		

Q2 Do the flowers power the buses' charging?

NOT SHOWN

EVIDENCE FOR THE CLAIM'S IMPRESSION

Modelled output is about 85 MWh a year [17], of the same order as a few hours a day of the shuttle's likely use [23–25]. The statement says “offset”, which a yearly balance through the grid can deliver.

EVIDENCE AGAINST

No capacity, metered output, bus consumption or wiring has been published, and no feasibility study was made [10] ♦. The new depot at the hub charges buses during the night [22], when the flowers are closed [4].

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: a part of the charging can be offset; how much has not been shown.

7

Testing the claim

Sub-claim	Said by	What the evidence shows	Rating
A. Fifteen solar flowers were installed at the Gozo Multi-Modal Hub, Ta' Xhajma	Statement [1–3]; Commission [4]	The maker reports fifteen SmartFlowers at the hub [5]; Parliament was told all were working [10] ♦.	ACCURATE
B. “The first of its kind in Europe”; “these first solar flowers in Europe”	Statement [1–3]; Camilleri, quoted [1, 2]; Commission [4]	SmartFlowers, an Austrian product, were in service in Spain (2015), Austria (2016) and the UK (2021) [11–15]; the maker calls Gozo's “one of the largest installations in Europe” [5].	CONTRADICTED
C. The first of its kind used in Malta; second in the world by number at one site	Camilleri, quoted [1, 2]	First in Malta: consistent with the reply to Parliament [10] ♦ and our search. Second in the world: not tested.	NOT CONTRADICTED
D. The energy will supply, or offset the charging of, the Park and Ride electric buses	Camilleri, quoted [3]; statement [2]	No capacity, metered output or bus use published; no feasibility study [10] ♦. Modelled output equals about 2–3 hours a day of the shuttle (indicative) [17, 23–25].	NOT SUBSTANTIATED
E. Each flower yields up to 40% more energy than traditional panels	Statement [3]; Commission [4]	PVGIS: 36.5% more than fixed panels at the best angle at the site [17]; studies 18–31% elsewhere [18]. Not measured here.	LARGELY ACCURATE
F. About €850,000, funded by the EU recovery plan (C1-I5)	Statement [1, 3]; Commission [4]	Parliament was told €699,612 excluding VAT [10] ♦; the likely award, €718,830, is €848,219 with VAT [26, 27]; C1-I5 funds solar panels in public spaces [28].	CONSISTENT

8

Verdict and requests for evidence

VERDICT

CONTRADICTED

The claim that Gozo's fifteen solar flowers are Europe's first is contradicted by dated records of the same product elsewhere in Europe since 2015. Confidence: high.

Why. (1) The headline claim, in the minister’s words, the statement and the Commission’s page, is that these are Europe’s first solar flowers, or its first installation of the kind. (2) The units are SmartFlowers, and independent, dated records show the same Austrian product in service in Spain, Austria and the UK years earlier, including a 2016 pilot by an Austrian motorway operator. (3) The maker calls Gozo’s installation one of the largest in Europe, not the first. The evidence points against the claim as worded: *Contradicted*. That the energy powers or offsets the buses’ charging

is *Not substantiated*: no figures show how much.

Why high confidence. Several independent lines agree: the maker’s own pages, local and trade reports in three countries, and an Austrian public-procurement record. **Why not Misleading.** The words are not a defensible statement that leaves a wrong impression; they state a fact that the records contradict.

What this verdict does not say. It does not say the flowers do not work, that the money was wasted, that the energy is not used at the hub, or that anyone meant to mislead. A statement closer to the evidence: *“Malta’s first solar flowers, one of the largest such installations in Europe, will offset part of the electricity used to charge the Park and Ride buses.”*

Evidence we are asking for

- ☐ What “first of its kind in Europe” referred to, and the source for “second in the world”.
- ☐ The installed capacity and model of the fifteen units, and their metered output since October 2025.
- ☐ The metered electricity used to charge the Park and Ride buses, their model and battery, the shuttle’s hours, and how the flowers are connected (to the chargers, the hub or the grid).
- ☐ The tender and contract documents (CT3013/2024, if that is this contract), and the reason the figure given to Parliament differs from the award notice.

RIGHT OF REPLY

A right of reply will be sought from the Ministry for Gozo and Planning, with the Public Works Department, before this check circulates beyond Miżien’s site, with a fixed deadline (suggested 14 days). The European Commission’s project page repeats the claim and may be offered the same. Responses will be appended and the verdict revisited. A *Contradicted* verdict is not circulated elsewhere before that deadline passes.

9

Limitations

- gov.mt and parliament.mt could not be read from our network; we used TVM News, Lovin Malta and Newsbook’s report of the answers in Parliament ♦.
- We did not establish which European SmartFlower was the first, only that several preceded Gozo’s, from news reports and dealer pages.
- The capacity of the Gozo units is assumed from the maker’s current 2.5 kWp model. PVGIS models an ideal two-axis tracker with a default 14% loss; the petal design, shading at the hub and downtime are not modelled.
- The bus-consumption figures (1.45–2.1 kWh per km) were modelled for other buses and routes and read as abstracts; the route length comes from car routing on OpenStreetMap. Figure 2B is an order of magnitude, not an estimate of the share covered. EU award notice 742398-2024 is matched to this contract by its subject, buyer, date and value; it names no site.

References

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Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** where a check finds a claim Not substantiated, Misleading or Contradicted (or a pledge Not measurable, Off track or Missed), before wider circulation, and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	6 Oct 2026	First issue. Pending right of reply (Ministry for Gozo and Planning; Public Works Department).